MILITARY SPECIFICATION

AMPLIFIER-POWER SUPPLY-RECEIVER AM-2310(*)/ASQ AND ANTENNA AS-909(*)/ARA-48

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements of Amplifier-Power Supply-Receiver AM-2310/ASQ and Antenna AS-909/ARA-48. When used as a part of the AN/ASQ-56, AN/ASQ-57, AN/ASQ-58 or other centrals this equipment shall provide the following:

(1) Reception of UHF voice communications.

(2) A continuous indication, in degrees, of the relative direction of arrival of received signals in the 200-400 mc range.

(3) UHF ADF amplifier and antenna switching functions.

(4) Generation of B- and bias voltages for use by other equipments which are a part of the integrated electronics centrals.

(5) Relay switching and distribution of AC and DC voltages for operation of the equipment in the integrated electronic centrals.

1.2 Associated Equipment - The equipment shall be capable of operating with the following associated equipment which is not to be supplied as part of this equipment.

Item	Type Designation
Receiver-Transmitter, Radio	RT-542/ASQ
Receiver-Transmitter, Radio	RT-541/ASQ
Decoder, Pulse	KY-309/ASQ
Coder-Receiver-Transmitter	KY-308/ASQ
Headset, Electrical	H-87A/U
Indicator, Bearing-Distance-Heading	ID-663/U
Control, Receiver	C-1457/ARR-40
Mounting-Plenum	MT-2243/ASQ-56 or MT-2244/ASQ
Antenna	AT-141/ARC
Indicator	ID-90/ARN-6, ID-250/ARN-14 or ID-416/ARN

FSC 5895

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5588	Enamel; Wrinkle Finish, For Aircraft Use
MIL-I-22668	Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57, AN/ASQ-58.
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

Federal

Fed. Std. No. 595	Colors (collection of 358 standard colors for paints and other materials)

2.2 Availability of Documents:

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, parts and materials requirements shall conform to Specification MIL-I-22668.

3.2 Design and Construction:

3.2.1 General - This equipment shall conform with the applicable requirements of Specification MIL-I-22668 for design, construction and workmanship except in the paragraphs as modified below:

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3.2.4.1 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.5.1 Internal and External Connectors - Connectors used for input wiring or interconnection of removable subassemblies shall be type DA manufactured by the Cannon Electric Company or an approved equivalent. External connectors shall be Amphenol Type 126-10753 or an approved equivalent in accordance with MIL-C-26482. The wires and terminals of this connector shall be potted with a sealing compound which is capable of withstanding temperatures up to 125°C.

3.2.8 Antenna Conducted CW - The antenna conducted interference shall be less than 300 microvolts in the frequency range of 200 to 400 mc when the guard receiver is operating and less than 100 microvolts when the main receiver is in use.

3.2.11 Standard Conditions - Cooling Air - 25 LB/HR of ambient room air.

3.2.12 Maximum Altitude - The maximum altitude under service conditions for the equipment shall be 75,000 feet.

3.2.12.1 Minimum External Cooling Requirements - The AM-2310/ASQ shall operate satisfactorily under the temperature-altitude conditions specified for Class 2 equipment per MIL-E-5400 when external cooling air is supplied at the rate specified in figure 1. The curve of figure 1 specifies the minimum cooling air flow rates that are required for air inlet temperature varying from 0°F to 120°F in a 168°F environment. The cooling air weight flow as a function of the Amp-PS-Rec equipment pressure drop is specified in the following equation:

$$W = \left[0.28\ (\sigma \Delta P)^{0.54}\right] \pm 10\%\ \text{lb/min}$$

W = Cooling air flow rate (lb/min)

$\sigma \Delta P$ = Corrected total pressure (inches H_2O)

σ = Density correction ρ/ρ_r

ρ_r = Standard density (0.0765 lb/ft^3)

3.3 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements apply under standard conditions.

3.3.1 Receiver Performance:

3.3.1.1 Frequency Characteristics - The receiver shall be capable of reception of 20 channels within the range of 265.0 to 284.9 mc, and one guard channel within the range of 238.0 to 248.0 mc as outlined below.

(1) By selection of a CR-23/U crystal of the proper frequency, channel one reception shall be obtained on any frequency within the range of 265.0 to 265.9 mc. Similarly, channel two shall be selected in the range 266.0 to 266.9 mc, channel 20 shall be selected in the range 284.0 to 284.9 mc. Unless otherwise specified, the assigned channel frequencies shall be as follows:

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COOLING AIR FLOW RATE
VS
COOLING AIR TEMPERATURE
A'4-2310/ASQ

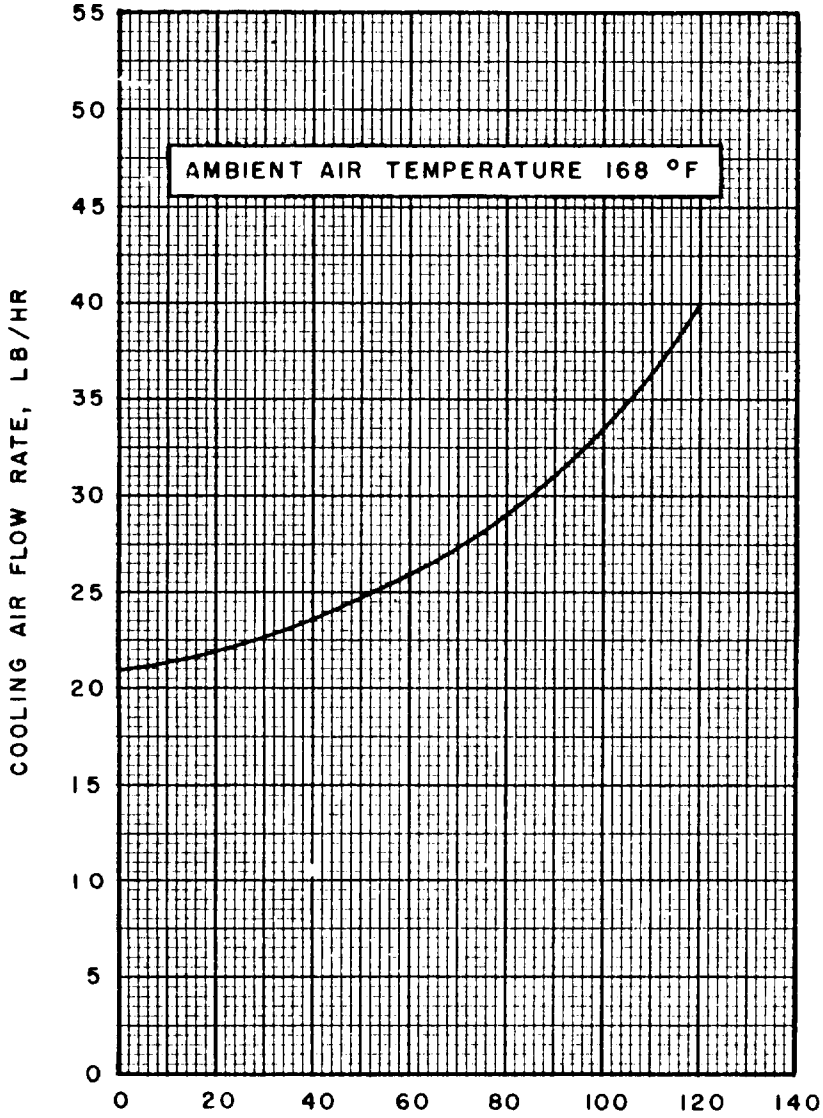


FIGURE 1

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1	265.2 mc	11	275.6 mc
2	266.6 mc	12	276.2 mc
3	267.6 mc	13	277.2 mc
4	268.6 mc	14	278.0 mc
5	269.8 mc	15	279.0 mc
6	270.8 mc	16	280.4 mc
7	271.6 mc	17	281.0 mc
8	272.2 mc	18	282.0 mc
9	273.2 mc	19	283.0 mc
10	274.8 mc	20	284.2 mc

(2) By selection of a CR-55/U crystal of the proper frequency, reception of one guard channel shall be obtained with the range 238.0 to 248.0 mc. Unless otherwise specified, guard channel operation shall be adjusted for 243.0 mc.

3.3.1.2 Sensitivity - The sensitivity of the receiver shall be such that an r-f input of 6 open circuit microvolts modulated 30% at 1000 cycles produces an audio output of at least 50 milliwatts with a signal plus noise-to-noise ratio (modulated to unmodulated) of 10 db or greater. The sensitivity (in microvolts for a 10-db signal plus noise-to-noise ratio) measured under service conditions shall not differ from that obtained under standard conditions by more than four to one, and shall not be worse than 18 microvolts at any frequency. Under the reduced performance condition, a signal plus noise-to-noise ratio of at least 10 db shall be obtained when a 10,000-microvolt signal modulated 30% at 1000 cps is applied.

3.3.1.3 AVC Characteristics - The AVC action shall be such that, when a 1000 microvolt signal modulated 30% at 1000 cps is applied, the audio output power shall be 250 milliwatts (-10%, +20%); and when the signal input (30% modulated at 1000 cps) is varied from 10 microvolts to 100,000 microvolts, the audio output power shall not vary by more than ±3 db from the audio output obtained with the 1000 microvolt signal. Additionally the receiver shall be protected against blocking under conditions of input signals up to 0.5 volt.

3.3.1.4 Selectivity - The over-all selectivity of the receiver shall fall within the limits set forth below for either standard or service conditions:

Attenuation Below Resonant Response	Total Receiver Bandwidth
6 db	60 kc min.
60 db	200 kc max.

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3.3.1.5 Images and Other Spurious Responses - When receiving on any channel, the receiver shall attenuate images and other spurious responses to the maximum practicable extent and, in any case, not less than 60 db relative to the desired signal. Not more than 3 spurious responses on a guard receiver channel shall be attenuated between 40 and 60 db.

3.3.1.6 I-F Rejection - The receiver shall present an attenuation of not less than 80 db relative to the response of the desired signal to signals on the frequencies of the intermediate frequency amplifiers.

3.3.1.7 Regeneration - The interstage and over-all regeneration of the receiver shall be so minimized that no evidence of unnecessary oscillation, instability, or excessive regeneration shall occur under either standard or service conditions.

3.3.1.8 Noise Peak Limiter - The second detector of the receiver (following the intermediate frequency amplifier) shall be provided with an instantaneous noise peak limiter. The noise limiting action shall effectively limit the audio output voltage of modulating surges or peaks exceeding 50% ± 15% modulation of r-f signal. Its action shall be rapid enough to prevent loss of intelligibility due to reduction in receiver gain under the influence of continuous r-f noises such as are caused by commutators, ignition systems, etc. When a pulsed signal having a pulse duration of 10 microseconds, a pulse repetition frequency of approximately 1000 pulses per second, and peak amplitudes up to 0.1 volt is applied on the same frequency as a desired signal modulated 30% at approximately 1000 cps, not more than 50 microvolts input (desired signal) shall be required to obtain a signal plus noise-to-noise ratio of 3 db.

3.3.1.9 Audio Characteristics - The audio characteristics of the receiver shall be as set forth below. Except where otherwise stated, these requirements apply under both standard and service conditions.

3.3.1.9.1 Capability - Under standard conditions, the receiver shall be capable of delivering an output of at least 250 milliwatts when a signal having any level between 100 and 100,000 microvolts modulated 30% at 1000 cps is applied, and the total harmonic content of the receiver output shall not exceed 10% at the 250 milliwatt level. Under service conditions, the output shall not be more than 6 db below that obtained under standard conditions. Under the reduced performance conditions, an output of at least 30 milliwatts shall be obtained when a 10,000 microvolt signal modulated at 30% at 1000 cps is applied.

3.3.1.9.2 Over-All Fidelity - Audio response at modulation frequencies from 300 to 4000 cps shall not vary more than +1 to -3 db relative to the response at 1000 cps. Modulation frequencies outside the 300 to 4000 cps range shall be attenuated the maximum practicable amount without employment of large and/or heavy filter components in the receiver.

3.3.1.9.3 Auxiliary Output - When a 1000 microvolt signal modulated 30% at 1000 cps is applied to the receiver, the auxiliary output shall be at least 0.25 volt, and when the modulation frequency is varied from 70 to 7000 cps, the output for any frequency in the range shall be within ± 3 db of the output obtained at 1000 cps. The phase shift through the receiver shall be such that this equipment is compatible with the automatic direction finding equipment.

3.3.1.10 Signal to Noise Type Squelch Characteristic - The receiver squelch shall be controlled by the audio signal plus noise-to-noise ratio and shall be used with maximum r-f gain. Under conditions of 6 microvolts r-f input modulated at 400 cps, the squelch control provided within the receiver shall have such a range as to permit squelch operation from no modulation to 30% modulation. With the squelch control adjusted to obtain normal audio output with modulation input required for 6 db signal plus noise-to-noise ratio, the audio shall be effectively suppressed when the modulation input is reduced to a 1 db signal plus noise to noise ratio. The squelch operation shall be smooth and positive and shall effect a change in audio output of 10 db or more at the threshold of operation. With the squelch adjusted as above the audio output shall be 50 milliwatts or greater with 6 microvolts input modulated 30% at 1000 cps.

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3.3.1.11 Noise Modulation - Under standard conditions, the signal plus noise-to-noise ratio shall be at least 30 db when a 10,000 microvolt signal modulated 30% at 1000 cps is applied and the receiver output (signal plus noise) is adjusted to 250 milliwatts.

3.3.1.12 Channel Changing Time - The time between the instant the channel selector switch is positioned to select a given channel, and the instant the equipment is in an operative condition on the selected channel shall not be greater than six seconds under service conditions. This requirement shall hold for shifting the operating frequency of the equipment between any two available channels. During channel changing operation, the output of the receiver (headset circuit) shall be shorted or otherwise reduced to zero.

3.3.1.13 Accuracy of Automatic Tuning - When the same frequency channel is selected several times in succession, the receiver audio power (with an input signal of 100 microvolts or greater modulated 30% at 1000 cps) shall not vary by more than 0.5 db from the power outputs obtained initially. This requirement shall be met under standard or service conditions.

3.3.1.14 Automatic Direction Finding - When installed with certain associated equipment the equipment shall be suitable for automatic direction finding. When used for this purpose, the receiver will receive radio-frequency signals from the direction finding antenna, amplify and demodulate them, and deliver the low-frequency modulation components to the direction finder amplifier and indicator equipment.

3.3.1.15 Frequency Stability - The frequency stability of the complete equipment after operation under either standard or service conditions for five minutes or more shall be such as to maintain the equipment frequencies within the limits shown below provided that the required quantity of cooling air is supplied.

Receiver resonant frequency tolerance: ± 22.5 kc.

3.3.1.16 Terminations:

3.3.1.16.1 Receiver R-F - R-F input to the receiver shall be applied through a 52-ohm cable from a signal generator having an effective source impedance of 52 ohms. Sensitivity shall be measured in terms of "open-circuit" microvolts.

3.3.1.16.2 Audio Terminations - The audio output (headset) of the receiver shall be terminated in a 300-ohm noninductive load which shall include means for measuring audio frequency power output. The auxiliary (high fidelity) output of the receiver shall be terminated in a resistance load of 20,000 ohms including the load presented by measuring devices.

3.3.2 Power Supply Performance:

3.3.2.1 Function - The power supply portion of AM-2310/ASQ shall provide B+ and bias voltages described in Table 1 to other equipments of a central. In addition, the power supply shall distribute 115 VAC, 27.5 VDC, and 26 VAC to other equipment.

3.3.2.1.1 Back-Up Provision - Means shall be provided within the equipment to furnish voltages to essential equipment in case of failure of the main supply. The switching to the back up condition shall be accomplished automatically. The switching shall be provided between the following voltage sources.

Main Supply Voltage	Back-Up Supply Voltage
430	275
130	130 from 275 volt bleeder

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3.3.2.2 Output Characteristics - Normal - The B+ and bias supplies shall have the following characteristics under normal operation. See Table 1.

3.3.2.3 Output Characteristics - Reduced Performance - The B+ and bias supplies shall have the following characteristics when operating in the reduced performance condition. See Table 2.

3.3.2.4 Main Power Relay - A main power relay shall be included within the equipment. This relay when energized externally, shall apply the following voltages to this equipment and to certain other equipment in the central:

27.5 VDC essential bus	115 VAC 3 phase essential bus
27.5 VDC primary bus	115 VAC 3 phase primary bus
26 VAC single phase bus	115 VAC single phase monitor bus

3.3.2.5 Time Delay - During normal operation 27.5 VDC shall be supplied to other equipments 45-125 seconds after the main power relay is energized. This voltage shall be supplied from the 27.5 VDC essential bus. The maximum load shall not exceed 0.5 amp. When the 27.5 VDC primary bus is not energized, there shall be no delay in application of 27.5 VDC to other equipments.

3.3.2.6 Sensing Relays - When the voltage levels of either the +430 VDC or +130 VDC supplies drop to values less than 10% of nominal, the equipment shall switch to the back-up condition described in 3.3.2.1.1. Failures of this type shall cause an external warning light to be activated from 27.5 VDC.

3.3.2.7 Power Input - Power dissipated within the AM-2310/ASQ shall not exceed 180 watts under normal conditions. The main connector and associated wiring shall be such that they are capable of handling and distributing total input currents which shall not exceed the following values:

115 V AC single phase monitor bus	5.0 amp
115 V AC 3 phase essential bus	2.5 amp
115 V AC 3 phase primary bus	1.0 amp
27.5 V DC essential bus	5.0 amp
27.5 V DC primary bus	5.0 amp
26 V AC single phase instrument bus	3.0 amp

3.3.3 ADF Performance

3.3.3.1 Frequency - The equipment shall operate over the frequency range of the RT-542/ASQ, 225-400 mc, utilizing a modulated or unmodulated r-f signal.

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POWER SUPPLY PORTION OF
AMPLIFIER-POWER SUPPLY-RECEIVER AM-2310/ASQ

OUTPUT CHARACTERISTICS

TABLE 1. NORMAL CONDITIONS

Nominal Values 115 Volts AC				Maximum Values 117.5 VAC Min. Load		Minimum Values (7) 108.5 VAC Max Load	
Nominal Voltage	Nominal Load Resistance Ohms	AC(1) Impedance	Max RMS Ripple (4)	Resistance for Minimum Load	Maximum Voltage (6)	Resistance for Maximum Load	Minimum Voltage (5)
430 (TACAN)	107500	4500	1.2	143300	495	86000	378
430 (COMM)	1520	1200	0.8	1520	468 (2)	1417	380 (2)
275	11000	1050 (3)	0.7 (3)	13750	295	9166	247
250	8330	2300	0.9	8340	274	5000	221
140	666	10	0.04	666	143	666	137
130	250	400	0.4	2166	140	230	117
-180	10000	5500	2.3	10000	-203	10000	-152

(1) Maximum AC impedance 20 to 20,000 cycles with nominal loads connected.

(2) Voltages shown during the ON period for a maximum duty cycle of 5 minutes ON, 10 minutes OFF.

(3) With a 3 uf capacitor across load.

(4) Maximum ripple voltages for supply with nominal load.

(5) Minimum voltage under service conditions and maximum load.

(6) Maximum voltage under service conditions and minimum load.

(7) Voltages are for simultaneous connections of loads on all supplies

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POWER SUPPLY PORTION OF
AMPLIFIER-POWER SUPPLY-RECEIVER AM-2310/ASQ

OUTPUT CHARACTERISTICS

TABLE 2. REDUCED PERFORMANCE

	Maximum Values 120 VAC Min. Load			Minimum Values (6) 106 VAC Max. Load	
Nominal Voltage	Reduced Performance Ripple (5)	Resistance for Minimum Load	Maximum Voltage (4)	Resistance for Maximum Load	Minimum Voltage (3)
430 (TACAN)	1.2	143300	505	86000	370
430 (COMM)	1.2	1520	476 (1)	1417	372 (1)
275	0.8 (2)	13750	302	9166	241
250	1.0	8340	278	5000	218
140	0.05	666	143	666	137
130	0.5	2166	142	230	114
-180	2.5	10000	-207	10000	-148

(1) Voltages shown during the ON period for a maximum duty cycle of 5 minutes ON, 10 minutes OFF.

(2) With a 3 uf capacitor across load.

(3) Minimum voltage under service conditions and maximum load.

(4) Maximum voltage under service conditions and minimum load.

(5) Maximum ripple voltages for supply with minimum load.

(6) DC voltages are for simultaneous connections of loads on all supplies.

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3.3.3.2 Range - The equipment shall be capable of taking bearings when operating in a vertically polarized r-f field in the frequency range of 225 to 400 mc with a signal plus-noise-to-noise ratio of 10 db field strength when modulated 30% at 1000 cps and applied to the RT-542/ASQ through a quarter wave antenna. Bearing indication shall be within the limits of this specification, except that at the minimum signal strength, the hunting of the indicator needle shall not exceed ± 2.5°.

3.3.3.3 Indication

3.3.3.3.1 Visual Indication - The equipment, when used in conjunction with the RT-542/ASQ, shall provide 360° bearing indications on Indicators ID-90/ARN-6, ID-250/ARN-14 or equivalent 400 cps synchro driven indicators.

3.3.3.3.2 Accuracy of Indications - The total error inherent in the equipment shall not exceed ± 2.5° over the entire frequency range and with any field strength above the minimum specified in 3.3.3.2. This shall apply to standard conditions. For service conditions the total error shall not exceed ± 5°. This error does not include errors caused by field distortion arising from installation of the equipment in a specific aircraft. However, the design of the equipment shall include provision for the correction of quadrantal errors as great as 35°.

3.3.3.3.3 Bearing Speed - The equipment shall be capable of taking a bearing on a signal at an average of at least 40° per second from any starting point between 50° and 179° under standard conditions. For starting points less than 50° but more than 10° the average speed shall be at least 15° per second. This includes the time for the pointer to reach its normal action when on the bearing. Under combined conditions of low temperature and low voltage, the speed shall be at least 25° per second.

3.3.3.3.4 Lag - As the aircraft is being rotated at any speed up to 3° per second and taking a bearing on an r-f signal, the indicated bearings shall not lag the true "on-course" bearings by more than 5°.

3.3.3.3.5 Damping - On taking a bearing while in level flight on a stationary course, the overshoot of the antenna assembly shall not exceed 5°, nor shall the continuous hunting of the indicator pointer exceed ± 1.5°.

3.3.4 Attenuation Characteristics, Coaxial Relay - The coaxial relay shall not introduce more than 1.5 db loss when measured with 3 foot lengths of RG-142/U cable connected to the COMM ANT Connector and to the COMM RCVR connector of this equipment.

3.4 Detail Requirements:

3.4.1 Amplifier-Power Supply-Receiver AM-2310/ASQ - The amplifier-power supply-receiver shall meet the following requirements:

3.4.1.1 Function - The receiver shall operate normally as an ADF receiver. In addition, it will provide auxiliary or emergency voice reception. The operating frequency range shall be 265.0 to 284.9 mc with an additional guard frequency of 243.0 mc available when the 265.0 mc to 284.9 mc channels are not in use. The power supply section shall generate the required B+ voltages under 600 volts for the associated equipment of 1.2 using the voltage sources of the aircraft supply. In addition, power and time delay relays shall be provided to provide the required input power switching functions required by the associated equipment.

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3.4.1.2 Form Factor - The amplifier-power supply-receiver shall be contained in a single unit capable of being directly mounted to a shockmounted equipment shelf which in turn is mounted in an aircraft. It shall be possible to remove the unit from the equipment shelf and to remove the dust cover using ordinary hand tools without requiring removal of any cable. The over-all dimensions shall not exceed 7-9/16 by 6-15/32 by 32-5/16 inches.

3.4.1.3 Weight - The weight of the amplifier-power supply-receiver, less mounting, shall not exceed 37.5 pounds.

3.4.1.4 Contents - The amplifier-power supply-receiver shall contain the UHF receiver, UHF ADF and power supply circuits in the following 12 modular assemblies.

(1) RF and Injection Assembly - This assembly will provide the necessary stages of r-f amplification, the first mixer, injection crystal oscillator and multiplier circuits. The operating frequency range shall be 265.0 mc to 284.9 mc and the first IF output shall be 20.75 mc.

(2) Mechanical Tuning Assembly - This assembly will permit remote positioning of the RF and injection assembly to any of its 20 preset channels.

(3) IF and Guard Amplifier Assembly - This assembly will provide the necessary 20.75 i-f amplification, the second mixer, second injection oscillator and multiplier, the necessary fixed-frequency guard-channel r-f amplification, first guard-channel mixer and first guard-channel oscillator and multiplier circuitry. The second IF output shall be 1.85 mc.

(4) 1.85 IF Amplifier Assembly - This assembly shall provide a selective filter, the necessary stages of IF amplification, detector, AVC and noise limiter circuitry.

(5) Audio Amplifier Assembly - This assembly shall provide the necessary stages of audio amplification, an auxiliary audio output circuit, and S+N/N squelch circuitry.

(6) Relay Assembly - This assembly shall provide power switching, emergency switching, receiver main or guard channel switching, and time delay switching circuits.

(7) Inductor Assembly - This modular subassembly shall contain the power transformer, filament transformer and two filter chokes.

(8) Fuse Assembly - This assembly shall contain fuses for overload protection. It will also contain several power supply filter components.

(9) Rectifier Assembly - This assembly shall contain the necessary power supply rectifier circuits.

(10) Regulator Assembly - This assembly shall contain the necessary voltage regulator circuitry.

(11) Amplifier, Electronic Control - The Electronic Control Amplifier shall contain all the circuitry necessary for utilizing the demodulated output of the associated receiver to provide the input power to the a-c motor in the antenna assembly.

(12) Relay Module - This assembly shall contain a relay-operated coaxial switch which shall enable cross-over switching of the r-f inputs from Antenna AS-909/ARA-48 and the communications antenna to the main uhf receiver

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and the auxiliary uhf receiver. The assembly shall also contain relays which will switch adf control voltages and select the audio input to the control amplifier from the main uhf receiver or the auxiliary uhf receiver. The relay assembly shall have four plug-in type r-f receptacles (which mate with I. P. C. type 22475 or an approved equivalent) for connection to the two antennas and the receiver and two subminiature receptacles (which mate with Cannon type DEF-95 or an approved equivalent) for connection to the amplifier. The VSWR introduced by the antenna relay shall not exceed 1. 3:1 in the 225-400 mc frequency range.

3.4.1.5 Controls - The following semifixed adjustable internal controls shall be accessible for adjustment when the dust cover is removed.

3.4.1.5.1 Audio Amplifier Module Controls:

(1) Audio amplifier gain.

(2) Auxiliary output circuit gain.

(3) S+N/N squelch threshold.

(4) Side tone audio gain (not used in this application).

(5) Guard audio gain (not used in this application).

(6) The adjustment screw provided for feedback control shall be readily accessible from the top of the module.

3.4.1.5.2 Regulator Module Controls:

(1) Output voltage.

3.4.1.6 Electrical Connection:

Reference Designation	Receptacle Type	Function
5A13J1	Amphenol 126-20753 (or an approved equivalent)	Power, control, and r-f connections

3.4.1.7 Test Connector - A test connector shall be provided on the front of the AM-2310/ASQ for use with AN/USA-11 test equipment. This connector shall be type Bendix Pygmy PTOOP-14 19SW or an approved equivalent in accordance with Specification MIL-C-26482. The following voltages shall be provided to the connector:

Function	Pin Letter
Aux. Receiver AVC	A
RT-542/ASQ AVC	B
UHF ADF Audio	C
Ground	D
130 VDC B+	E
275 VDC B+	F
Aux. Receiver 130 DC B+	G
140 VDC B+	H
-240 VDC B+	J

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Function	Pin Letter
Aux. Receiver Audio	K
UHF ADF Motor Control	L
Aux. Receiver Tripler Drive Voltage	M
Aux. Receiver Guard Oscillator Drive	N
Aux. Receiver Mixer Oscillator Output	P
430 VDC B+	S
27.5 VDC Delayed	U

3.4.1.8 External Finish - The equipment covered by this specification shall be painted with color No. 2615 of FED-STD-595. The bottom surface of the equipment shall not be painted.

3.4.2 Antenna AS-909/ARA-48

3.4.2.1 Function - The antenna shall receive modulated or unmodulated transmissions in the 225 to 400 mc frequency band and provide an r-f input to the RT-542/ASQ. The antenna shall be such that its field pattern is essentially a cardioid. By switching the terminations of the receiving element in the antenna and providing a means of positioning the antenna to the crossover point, bearing information shall be obtained from the received signals.

3.4.2.2 Form Factor and Mounting - The maximum dimensions of the antenna shall not exceed 10-3/4 by 10-3/4 by 3-3/4 inches, not including mounting flanges. The antenna shall be capable of being flush mounted in the underside of the fuselage along the center line of the aircraft. If a flat surface of sufficient area is not available for the mounting of the antenna, it shall be possible to mount the antenna on a recessed ring, with no appreciable detriment to its operation, provided that the amount of recessing does not exceed 3 inches measured from the center line of the fuselage. The rotating element shall be kept at the same r-f potential as the housing by means of either spring contacts or an r-f choke, so that the fuselage of the aircraft will provide a suitable ground plane.

3.4.2.3 Weight - The total installed weight of the antenna assembly shall not exceed 9.5 pounds.

3.4.2.4 Contents - The antenna shall contain the following:

(1) The receiving element, backed up by its cavity.

(2) 115 V 400 cps motor for positioning the element

(3) R-F rotating joint

(4) High speed coaxial switch (chopper)

(5) Synchro transmitter

(6) Rate generator

(7) Appropriate gears

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3.4.2.5 Antenna Characteristics - The field pattern of the antenna shall be approximately a cardioid and by switching the termination of the feed, the mirror image of the pattern shall be obtained. The antenna shall be capable of receiving vertically polarized signals over the stated frequency range with a VSWR of less than 3:1 into a 50 ohm line. The gain of the antenna shall not be less than 21 db down from a standard quarter-wave stub (AT-141/AR) over the entire frequency band.

3.4.2.6 Component Characteristics

3.4.2.6.1 A-C Motor - The a-c motor shall be capable of rotating the antenna to the cross-over point at a speed necessary to meet the requirements stated in 3.3.3.3.3.

3.4.2.6.2 R-F Rotating Joint - The VSWR of 3:1 shall be an over-all figure and include the performance of the antenna cavity, r-f joint and high speed coaxial switch.

3.4.2.6.3 Coaxial Switch - The ratio of VSWR measured from either side of the high speed coaxial switch shall not exceed 1.3:1.

3.4.2.7 Electrical Connections - The antenna assembly shall have the following connectors:

(1) One I.P.C 82600 Type SC or an approved equivalent for the antenna transmission line.

(2) One Bendix Type PT-02A-16-26P or an approved equivalent for the low frequency connections.

3.4.2.8 External Finish - The AS-909/ARA-48 shall be finished with Black Wrinkle Enamel, Type I, per Specification MIL-E-5558 except for the dielectric cover which shall not be painted.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The equipment shall be subjected to the test requirements of Specification MIL-I-22668 to determine compliance with all the applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-I-22668.

6. NOTES

6.1 Use - The AM-2310/ASQ is mainly intended to be used as part of the Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57, or AN/ASQ-58. In these applications, it furnishes the required B+ power under 600 V for the electronic central, provides the auxiliary receiver function and mounting provision for the UHF ADF modules, and distributes the AC and DC supply voltages to other equipment of the central. The AS-909/ARA-48 provides the pilot with a continuous indication, in degrees, of the relative direction of arrival of received signals in the 225-400 mc frequency range.

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6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) Central Specification - Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57 and AN/ASQ-58 Specification MIL-I-22668 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 of Specification MIL-I-22668. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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